

Personal Loan Project

Post Graduate Program in Artificial Intelligence and Machine Learning

Prepared by Christian Caceres

January 14, 2026

Contents / Agenda

- Executive Summary
- EDA Results
- Data Preprocessing
- Model Building
- Model Performance
- Decision Rules and Feature Importance

Executive Summary

This analysis evaluates customer demographics, financial behavior, and relationship indicators to identify statistically significant drivers of personal loan acceptance and to inform feature selection and model design. The objective is to develop a decision tree model that enables AllLife Bank to identify customers with a higher propensity to accept a personal loan and to increase overall campaign effectiveness.

Model Evaluation:

Model 1 shows perfect training performance with 100% metrics, indicating a high risk of overfitting driven by a very deep, highly branched decision tree. Although test results remain strong—only 21 misclassifications out of 1,500 with low false positive (0.73%) and false negative (0.67%) rates—and key drivers such as Income, CCAvg, and higher Education levels are consistent, the model's complexity increases the risk of high variance, sensitivity to outliers, and weak long-term generalization, making stronger pruning necessary before deployment.

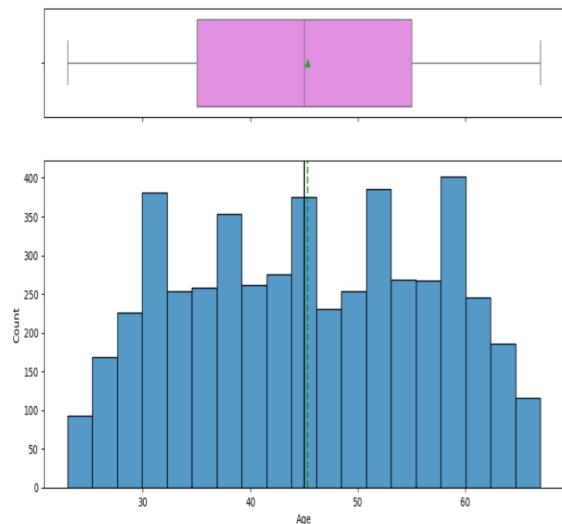
Model 2 shows clear underfitting due to an overly shallow, heavily pruned decision tree, resulting in poor training and test performance and very low precision (0.31). The model is strongly biased toward predicting loan acceptance, with high false positive rates ($\approx 20\text{--}22\%$) and limited ability to capture complex patterns, indicating that reduced pruning or increased depth is required before the model can be considered viable.

Model 3 emerges as the optimal solution, achieving a strong balance between performance and interpretability with 93% accuracy, very high recall (98–99%), and a solid F1-score of 74–76% on both training and test data. With an optimal $\text{ccp_alpha} \approx 0.010$, Model 3 generalizes well to unseen data while remaining concise and actionable, making it the most suitable candidate for deployment, particularly when the business priority is maximizing recall and overall campaign reach.

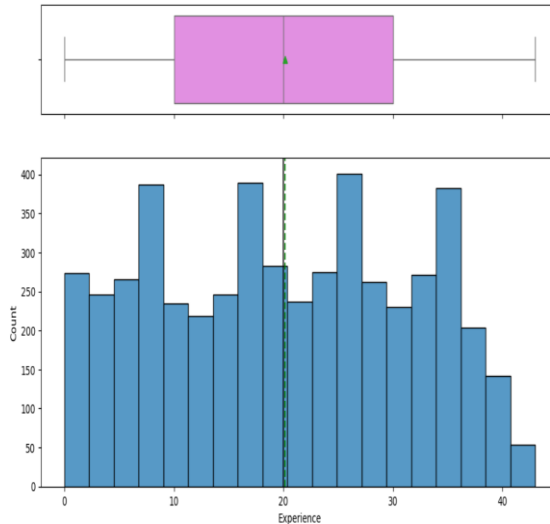
In conclusion, this analysis confirms that **income, education level, family size, and credit card spending** are the most influential drivers of personal loan acceptance and should remain central to AllLife Bank's customer targeting strategy. **Model 3** delivers the strongest balance of predictive accuracy, interpretability, and generalization, achieving high accuracy and recall within a concise and stable decision tree. To sustain performance and further reduce false positives, ongoing post-deployment monitoring and periodic model retraining are recommended as customer behaviors and market conditions evolve.

EDA Results: Age, Experience, Income

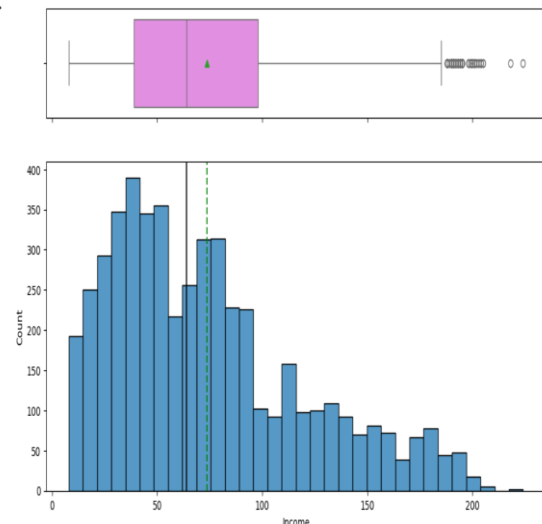
Age



Experience



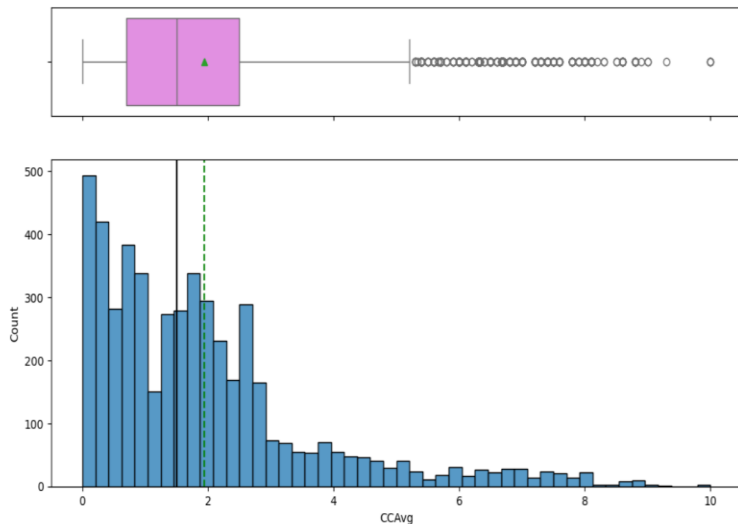
Income



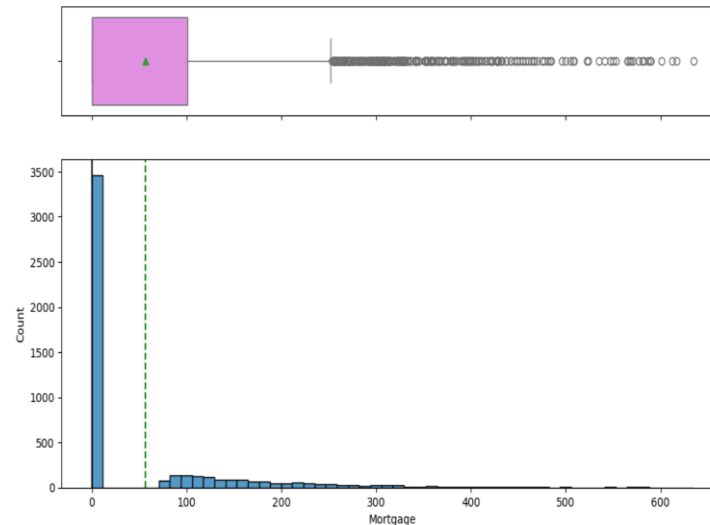
- Age is evenly distributed and stable, making it unlikely to distort analytical results or model performance. Both the median and mode are centered around approximately 45 years, and no significant outliers are present.
- Experience is unlikely to introduce instability into the model, as it is well distributed with both the median and mode centered around approximately 20 years. No significant outliers are observed.
- The distribution is right-skewed, with a long tail toward higher income levels. The presence of high-income outliers may affect model sensitivity.

EDA Results: CCAvg, Mortgage

Credit Card Avg



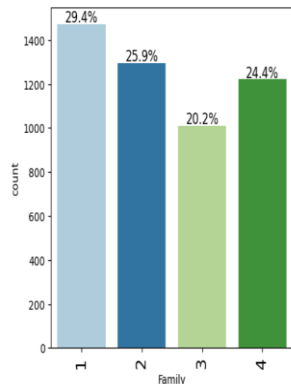
Mortgage



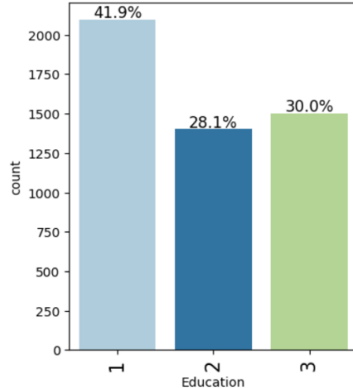
- CCAvg is strongly right-skewed. The majority of customers have low average credit card spending. The boxplot shows a significant number of outliers.
- Mortgage values are extremely right-skewed. The large spike at zero indicates many customers don't have a mortgage. The boxplot shows a significant number of outliers.

EDA Results: Family, Education, Securities, CD

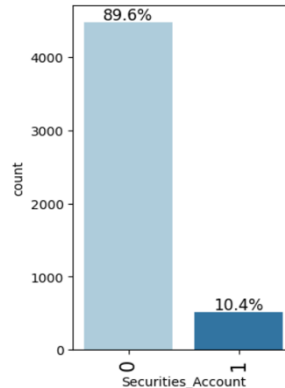
Family



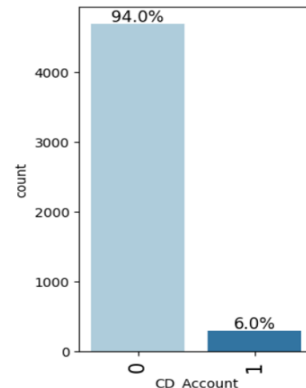
Education



Securities Account



CD Account



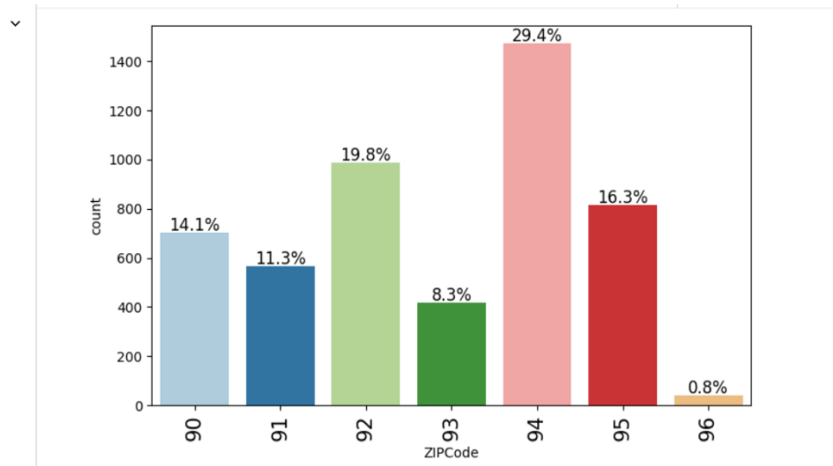
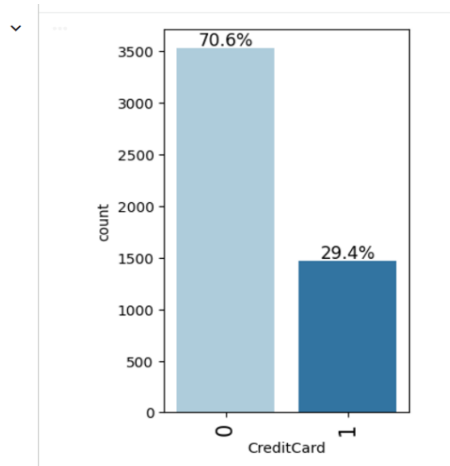
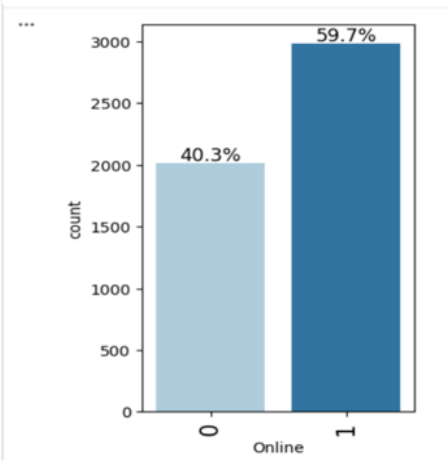
- Family exhibits a well-balanced distribution across family sizes - good candidate feature.
- Education shows a well-balanced distribution across education levels - strong predictor potential.
- Securities Account displays a highly imbalanced distribution – risk of overfitting the model.
- CD Account displays a highly imbalanced distribution – risk of overfitting the model.

EDA Results: Online, Credit Card, ZIP Code

Online

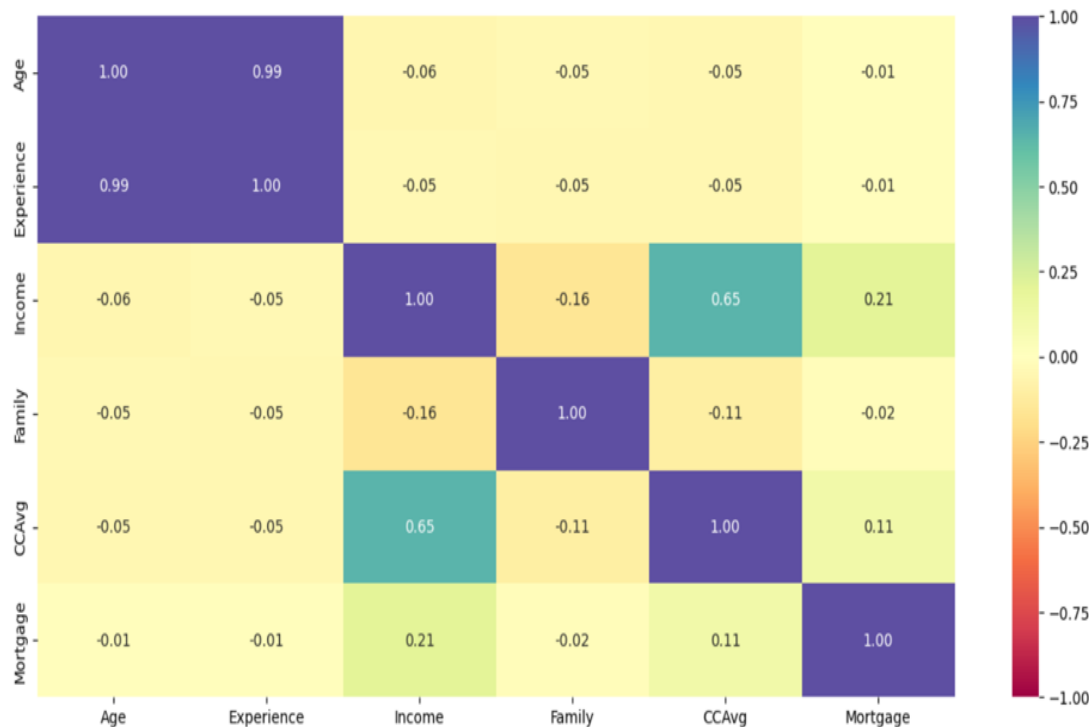
Credit Card

ZIP Code



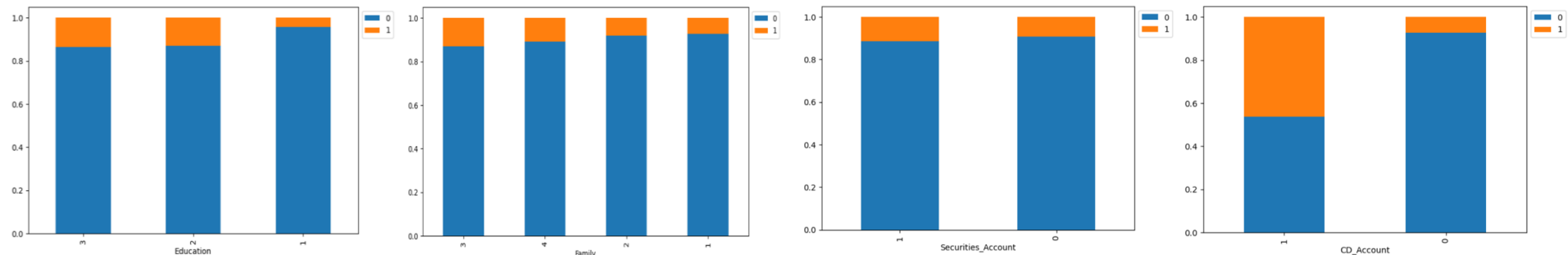
- Majority of customers (59%) use online banking.
- Majority of customer (70%) do not own a credit card.
- Zip 92 (19%) and Zip 94 (29%) account for ~48% of the total customer base.
- Further evaluate the potential benefits of including ZIP Code and Credit Card as features in the decision tree model, given their uneven distributions and the associated risk of overfitting.

EDA Results: Correlation

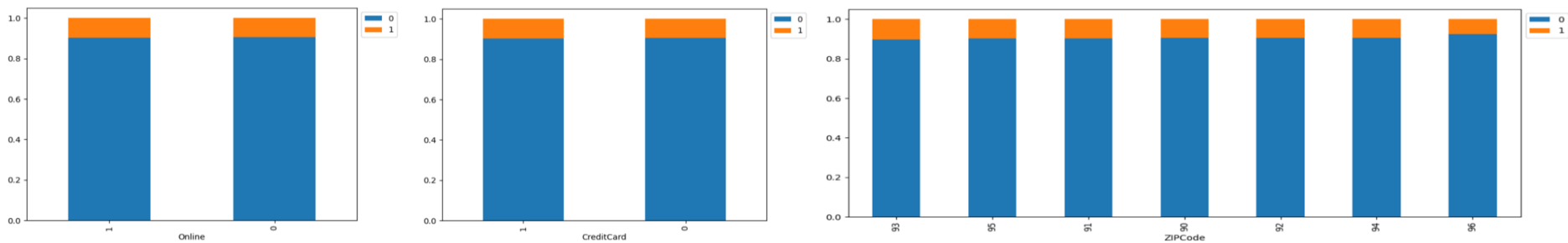


- High positive correlation (0.99) across 'Age' and 'Experience'. It is recommended to drop one variable to avoid multicollinearity.
- Strong positive correlation (0.65) across 'CCAvg' and 'Income'. This relationship warrants further analysis, supported by visualization through a stacked bar plot and a density curve.
- Moderate positive correlation (0.21) across 'Income' and 'Mortgage'

EDA Results: Personal Loan vs Education, Family, Securities, CD, Online, Credit Card, & ZIP Code

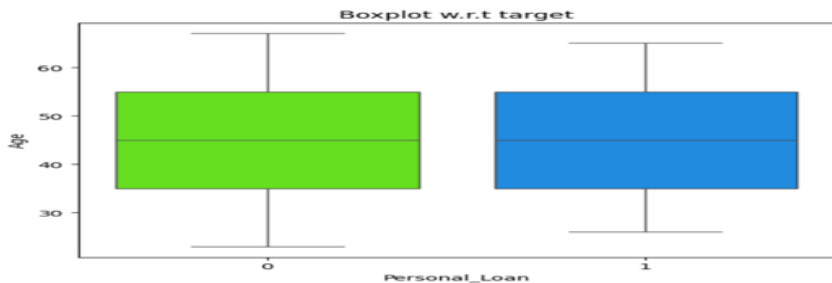
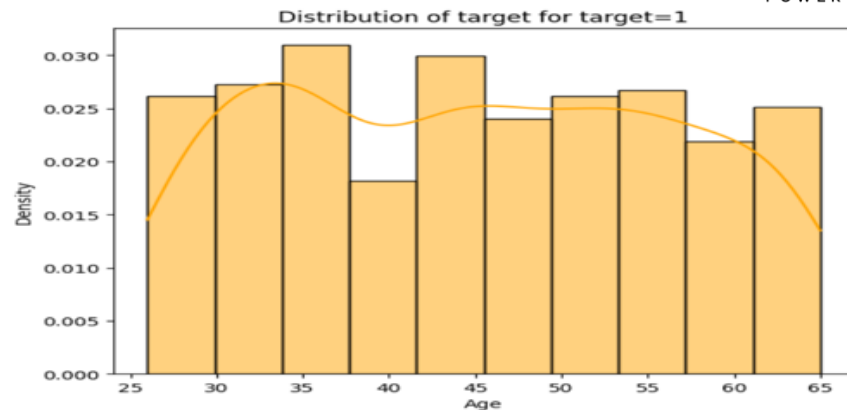
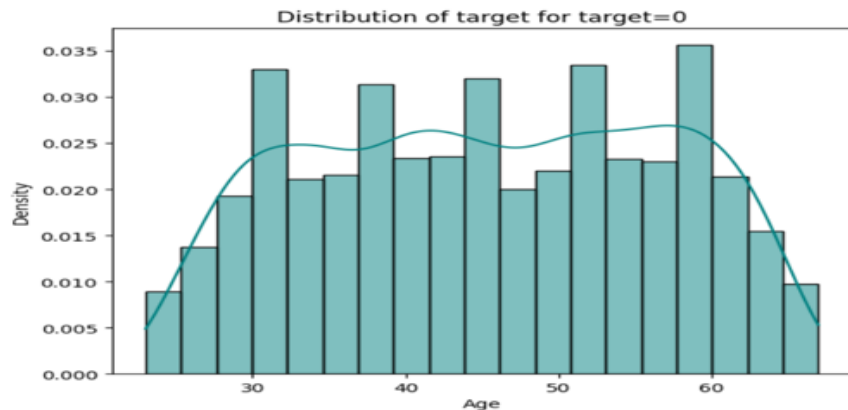


- The likelihood of personal loan acceptance is higher among customers with CD accounts, securities accounts, family sizes of three or more, and graduate or advanced education levels.



- Online banking usage and credit card ownership are not meaningful drivers of personal loan acceptance.
- Personal loan acceptance rates do not vary significantly at the zip code level.

EDA Results: Personal Loan vs Age



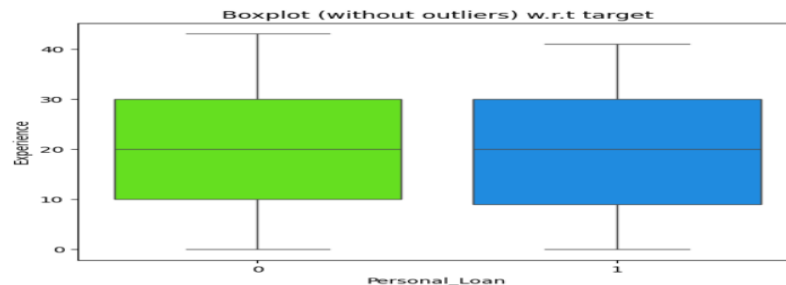
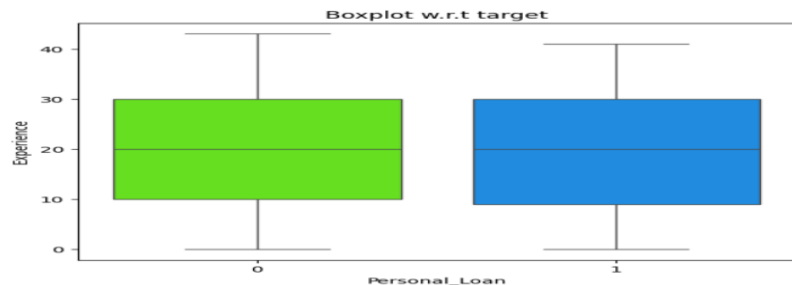
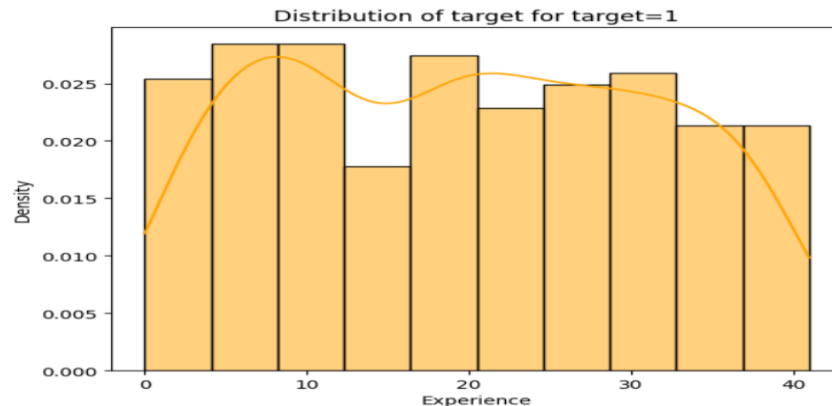
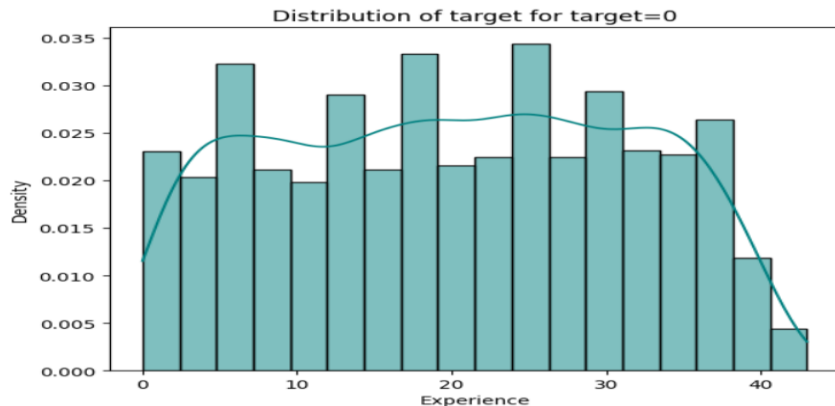
Didn't accept personal loan → target = 0

- No clear age cutoff that helps separate personal loan acceptors from non-acceptors.
- No strong concentration around a specific age group.

Accepted personal loan → target = 1

- No clear age cutoff that helps separate personal loan acceptors from non-acceptors.
- Customer density is slightly higher in the 30–50 age range.

EDA Results: Personal Loan vs Experience



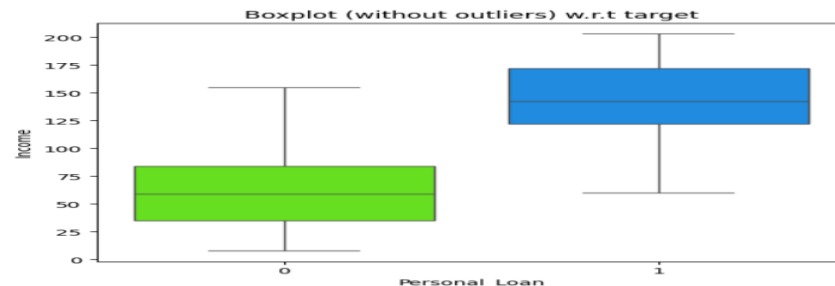
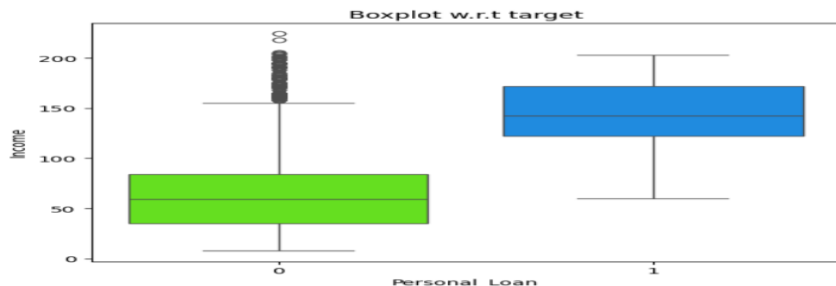
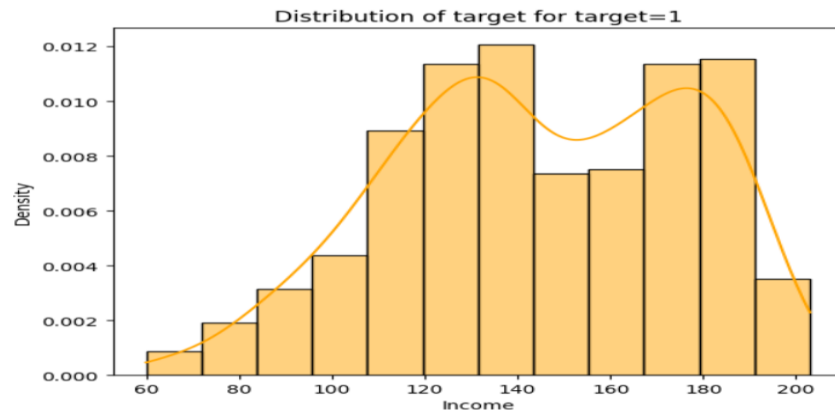
Didn't accept personal loan → target = 0

- Customers with less than 10 years of experience are less likely to accept a personal loan.
- Strong right-skewed distribution indicates a high concentration of early career professionals.
- Boxplot shows no outliers.

Accepted personal loan → target = 1

- Customers with 10 to 20 years of experience are more likely to accept a personal loan.
- Experience distribution is symmetric with a strong concentration of middle to high tenured career professionals
- Boxplot shows no outliers.

EDA Results: Personal Loan vs Income



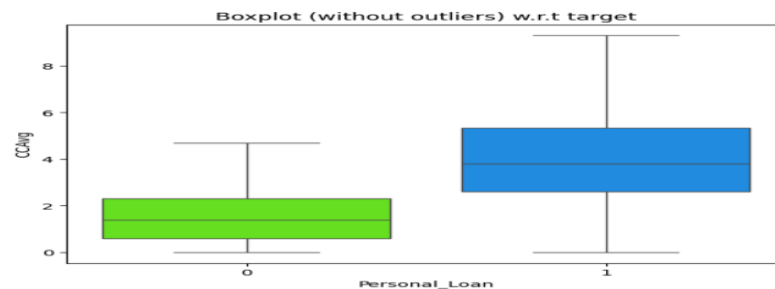
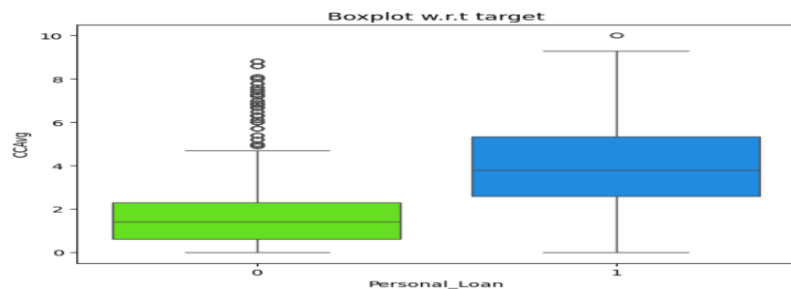
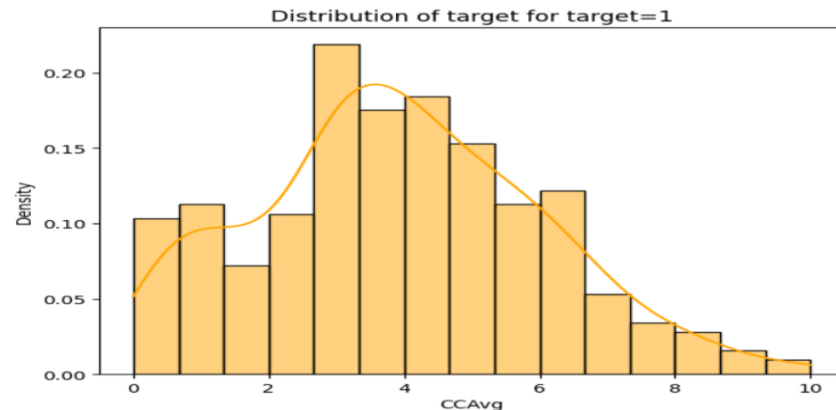
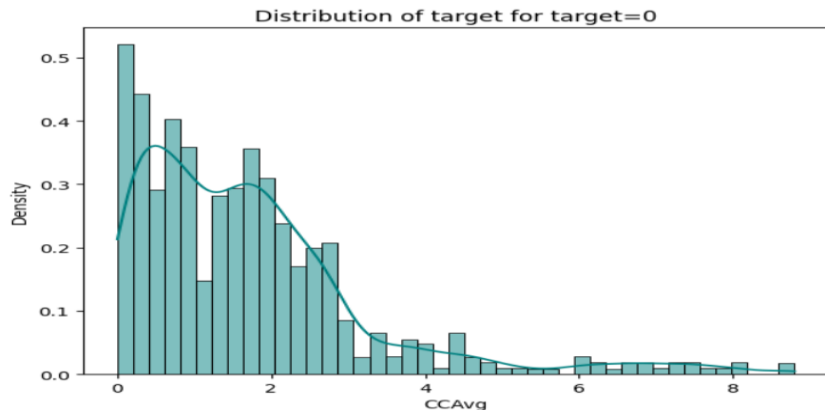
Didn't accept personal loan → target = 0

- Customers with low to medium income (\$40K - \$90K) are less likely to accept a personal loan.
- Right-skewed distribution with most values between \$10K and \$75K
- Boxplot shows a strong presence of high value outliers.

Accepted personal loan → target = 1

- Customers with middle to high income (\$120K - \$200K) are more likely to accept a personal loan.
- Income distribution is consistent with very few individuals earning less than \$100K
- Boxplot shows no outliers

EDA Results: Personal Loan vs CCAvg



Didn't accept personal loan → target = 0

- Customers with low CCAvg balances are less likely to accept a personal loan, as their limited credit card usage suggests lower familiarity with credit-base products.
- Strong right-skewed distribution with most values between \$0 & \$2K (Monthly CCAvg spending) – Boxplot shows a strong presence of high value outliers.

Accepted personal loan → target =1

- Customers with high CCAvg balances are more likely to accept a personal loan.
- Spending distribution is more consistent and less skewed – distribution is shifted to the right with most values between \$4K & \$8K (Monthly CCAvg spending) – Boxplot shows no outliers.

Data Preprocessing: Data Types

```
*** <class 'pandas.core.frame.DataFrame'>
RangeIndex: 5000 entries, 0 to 4999
Data columns (total 14 columns):
 #   Column                Non-Null Count  Dtype
---  -
 0   ID                    5000 non-null   int64
 1   Age                  5000 non-null   int64
 2   Experience            5000 non-null   int64
 3   Income               5000 non-null   int64
 4   ZIPCode              5000 non-null   int64
 5   Family               5000 non-null   int64
 6   CCAvg                5000 non-null   float64
 7   Education            5000 non-null   int64
 8   Mortgage             5000 non-null   int64
 9   Personal_Loan         5000 non-null   int64
10   Securities_Account    5000 non-null   int64
11   CD_Account           5000 non-null   int64
12   Online               5000 non-null   int64
13   CreditCard           5000 non-null   int64
dtypes: float64(1), int64(13)
memory usage: 547.0 KB
```

Data Types

- Int 64: 14 columns
- Float 64: 1 column
- No Missing Values
- Columns: 14
- Rows: 5000
- ID column should be dropped
- Target variable: 'Personal_Loan'

Data Preprocessing: Outlier Detection

Feature	% Outliers *	Interpretation
Age	0.00	No Outliers
Experience	0.00	No Outliers
Income	1.92	Minor Presence of Outliers
Family	0.00	No Outliers
CCAvg	6.48	High Presence of Outliers
Mortgage	5.82	High Presence of Outliers

*Percentage of outliers in each column using the IQR Method: $IQR = Q3$ (75th percentile) - $Q1$ (25th percentile)

Lower Bound: $Q1 - 1.5 \times IQR$

Upper Bound: $Q3 + 1.5 \times IQR$

Data Preprocessing: Anomalous Values

Checking for Anomalous Values

A `data["Experience"].unique()`

```
... array([ 1, 19, 15,  9,  8, 13, 27, 24, 10, 39,  5, 23, 32, 41, 30, 14, 18,
          21, 28, 31, 11, 16, 20, 35,  6, 25,  7, 12, 26, 37, 17,  2, 36, 29,
           3, 22, -1, 34,  0, 38, 40, 33,  4, -2, 42, -3, 43])
```

B `# checking for experience < 0`
`data[data["Experience"] < 0]["Experience"].unique()`

```
array([-1, -2, -3])
```

C `# Correcting the experience values`
`data["Experience"].replace(-1, 1, inplace=True)`
`data["Experience"].replace(-2, 2, inplace=True)`
`data["Experience"].replace(-3, 3, inplace=True)`

- 
- A** Variable: 'Experience' shows 47 unique values
 - B** There are 3 negative unique variables within this variable (-1, -2, -3)
 - C** As a best practice these negative values were replaced with positive values

Data Preprocessing: Feature Engineering

Feature Engineering

A # checking the number of uniques in the zip code
data["ZIPCode"].nunique()

7

B data["ZIPCode"] = data["ZIPCode"].astype(str)
print(
 "Number of unique values if we take first two digits of ZIPCode: ",
 data["ZIPCode"].str[0:2].nunique(),
)
data["ZIPCode"] = data["ZIPCode"].str[0:2]

data["ZIPCode"] = data["ZIPCode"].astype("category")

Number of unique values if we take first two digits of ZIPCode: 7

C ## Converting the data type of categorical features to 'category'
cat_cols = [
 "Education",
 "Personal_Loan",
 "Securities_Account",
 "CD_Account",
 "Online",
 "CreditCard",
 "ZIPCode",
)
data[cat_cols] = data[cat_cols].astype("category")

- 
- A** Variable: 'ZIPCode' shows 7 unique values
 - B** Shortened the length of the variable 'ZipCode' by selecting the first 2 digits
 - C** Converted the datatype for 7 variables from int64 to categorical since these variables are categorical not continuous

Data Preprocessing: Preprocessing for Modeling

Data Preparation for Modeling

A # dropping Experience as it is perfectly correlated with Age

```
X = data.drop(["Personal_Loan", "Experience"], axis=1)
Y = data["Personal_Loan"]

X = pd.get_dummies(X, columns=["ZIPCode", "Education"], drop_first=True)

X = X.astype(float)
```

B # Splitting data in train and test sets

```
X_train, X_test, y_train, y_test = train_test_split(
    X, Y, test_size=0.30, random_state=1
)
```

▶ print("Shape of Training set : ", X_train.shape)
print("Shape of test set : ", X_test.shape)
print("Percentage of classes in training set:")
print(y_train.value_counts(normalize=True))
print("Percentage of classes in test set:")
print(y_test.value_counts(normalize=True))

B Shape of Training set : (3500, 17)
Shape of test set : (1500, 17)
Percentage of classes in training set:
Personal_Loan
0 0.905429
1 0.094571
Name: proportion, dtype: float64
Percentage of classes in test set:
Personal_Loan
0 0.900667
1 0.099333
Name: proportion, dtype: float64

A Dropped variable 'Experience' since it is perfectly correlated to the variable 'Age' (Correlation: 0.99)

B Split data randomly into training data (70%) and test data (30%)
training set (3500, 17) -> used to train the model
test set (1500, 17) -> used to evaluate the model

- This is a standard step in ML to evaluate model performance on unseen data

Model Building: Approach

1

Define the classification model performance metrics using scikit-learn

```
def model_performance_classification_sklearn(model, predictors, target):  
    """  
    Function to compute different metrics to check classification model performance  
    """  
  
    model: classifier  
    predictors: independent variables  
    target: dependent variable  
    """  
  
    # predicting using the independent variables  
    pred = model.predict(predictors)  
  
    acc = accuracy_score(target, pred) # to compute Accuracy  
    recall = recall_score(target, pred) # to compute Recall  
    precision = precision_score(target, pred) # to compute Precision  
    f1 = f1_score(target, pred) # to compute F1-score  
  
    # creating a dataframe of metrics  
    df_perf = pd.DataFrame(  
        [{"Accuracy": acc, "Recall": recall, "Precision": precision, "F1": f1}],  
        index=[0],  
    )  
  
    return df_perf
```

- Accuracy: Overall correctness of the model.
- Recall: How well the model finds actual positives.
- Precision: How reliable the positive predictions are.
- F1: Balance between precision and recall.

2

Create a confusion matrix using scikit-learn to evaluate the classification model performance (correct predictions vs classification errors).

```
def confusion_matrix_sklearn(model, predictors, target):  
    """  
    To plot the confusion_matrix with percentages  
    """  
  
    model: classifier  
    predictors: independent variables  
    target: dependent variable  
    """  
  
    y_pred = model.predict(predictors)  
    cm = confusion_matrix(target, y_pred)  
    labels = np.asarray(  
        [  
            ["{0:0.0f}".format(item) + "\n(0.2X)".format(item / cm.flatten().sum())]  
            for item in cm.flatten()  
        ]  
    ).reshape(2, 2)  
  
    plt.figure(figsize=(6, 4))  
    sns.heatmap(cm, annot=labels, fmt="")  
    plt.ylabel("True label")  
    plt.xlabel("Predicted label")
```

- True Positives: Correct positive predictions.
- True Negatives: Correct negative predictions.
- False Positives: Incorrect positive predictions.
- False Negatives: Incorrect negative predictions.

3

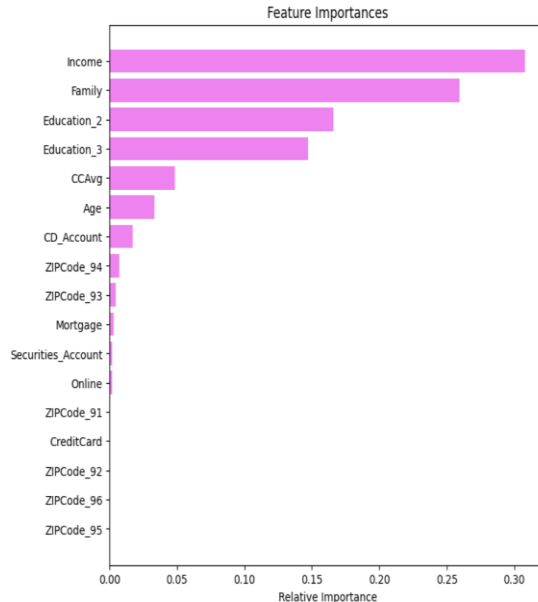
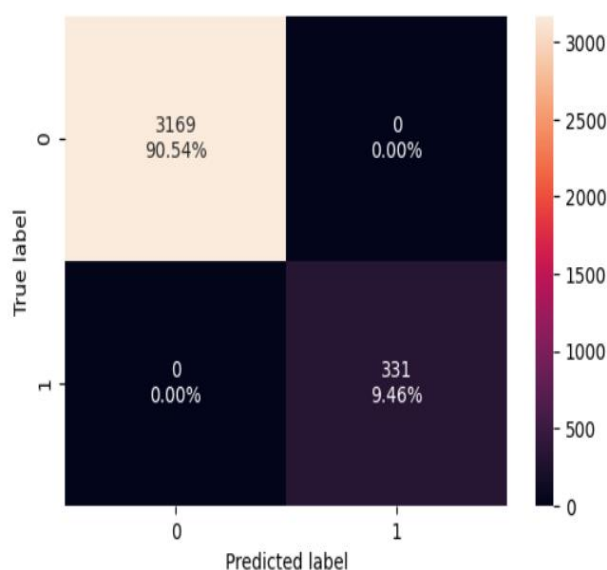
Create a decision tree using scikit-learn default.

```
model = DecisionTreeClassifier(criterion="gini", random_state=1)  
model.fit(X_train, y_train)  
  
...  
DecisionTreeClassifier  
DecisionTreeClassifier(random_state=1)
```

- Sklearn default decision tree without any hyperparameters
- Default setting: 'gini impurity' parameter

Model Building I: Performance (Training Data)

Decision Tree (Sklearn default)



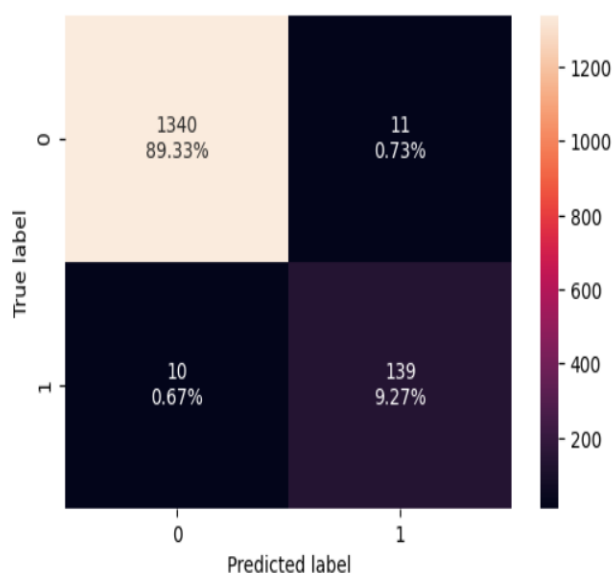
Performance Metrics

- **Accuracy: 1.0**
- **Recall: 1.0**
- **Precision: 1.0**
- **F1: 1.0**

- This confusion matrix shows perfect classification: False Positives: 0 & False Negatives: 0.
- The perfect training performance suggests a high likelihood of overfitting, indicating that the model may be fitting noise in the training set instead of capturing generalizable relationships.
- Performance metrics are all at 100%.
- Key predictors of loan acceptance include: (1) Income, (2) Family Size, (3) Education Level (Graduate) and (4) Education Level (Advanced).

Model Building I: Performance (Test Data)

Decision Tree (Sklearn default)



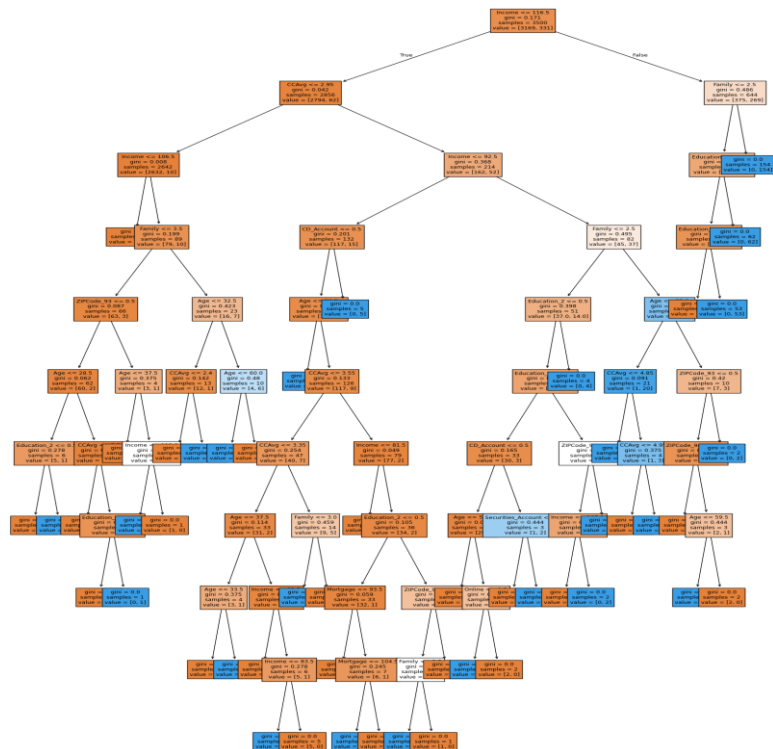
Performance Metrics

- **Accuracy: 0.98**
- **Recall: 0.93**
- **Precision: 0.92**
- **F1: 0.92**

- This confusion matrix performs well with unseen data, with only 21 misclassifications out of 1500 observations: False Positives: 0.73% & False Negatives: 0.67%
- Test data shows strong generalization with high precision and recall for loan acceptance.
- The model performs extremely well and has been able to learned meaningful patterns in the data while minimizing any noise.

Model Building I: Performance (Training Data)

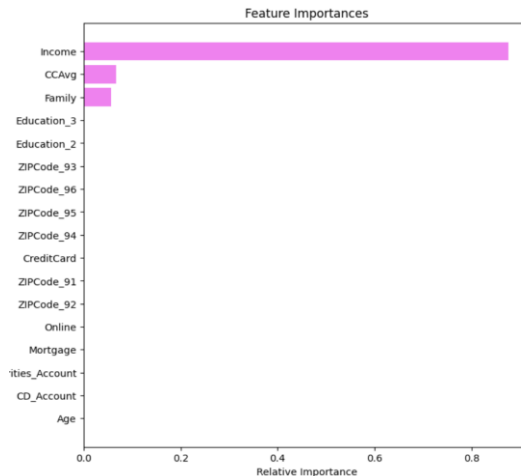
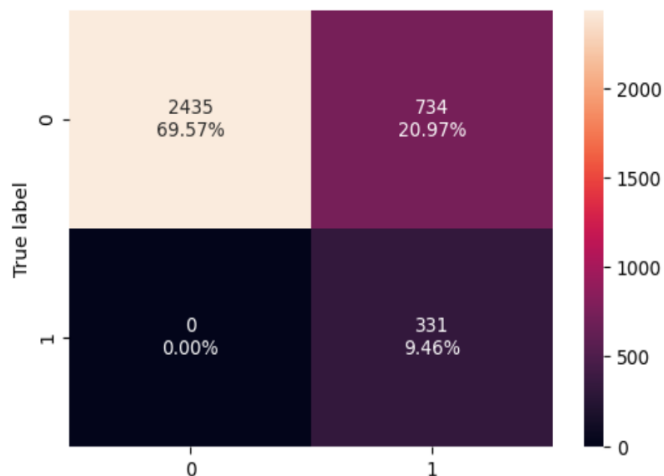
Decision Tree (sklearn default)



- The decision tree is very deep and highly branched
- Primary drivers include: (1) Income, (2) CCAvg, and (3) Education
- There is a high likelihood of overfitting
- Excessive depth suggest the need for stronger pruning
- Potential problems with highly branched trees include:
 - (a) Overfitting
 - (b) High Variance
 - (c) Sensitive to Outliers
 - (d) Poor Generalization
 - (e) Leaf Nodes with very few observations

Model Building II: Performance (Training Data)

Decision Tree (Pre-Pruning)



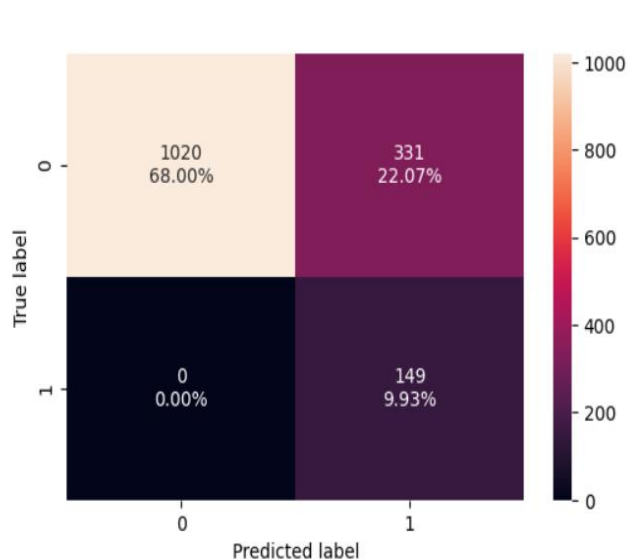
Performance Metrics

- **Accuracy: 0.79**
- **Recall: 1.0**
- **Precision: 0.31**
- **F1: 0.47**

- This confusion matrix suggests reduced performance with 734 misclassifications out of 3500 observations.
- The stark imbalance between false positives (20%) and false negatives (0%) indicates a strong bias in the model toward predicting the positive class.
- Low precision levels (0.31) indicate the model lacks reliability and poor performance.

Model Building II: Performance (Test Data)

Decision Tree (Pre-Pruning)



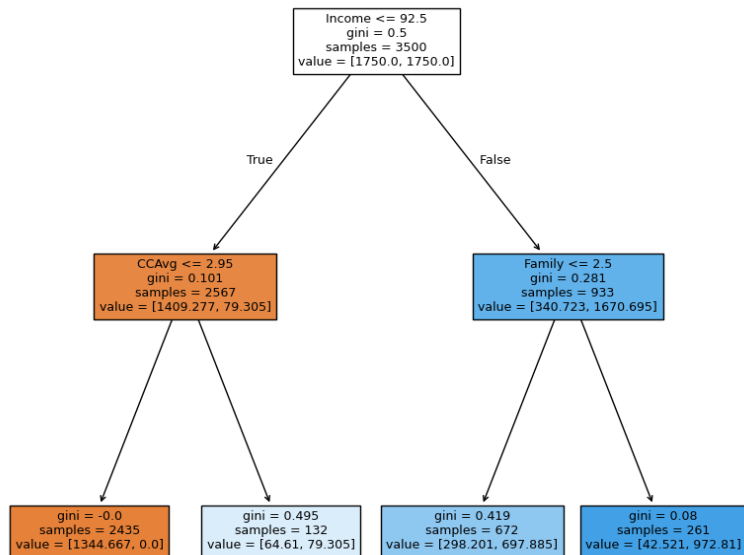
Performance Metrics

- **Accuracy: 0.77**
- **Recall: 1.00**
- **Precision: 0.31**
- **F1: 0.47**

- This confusion matrix suggests reduced performance on unseen data, with 331 misclassifications out of 1500 observations.
- The stark imbalance between false positives (22%) and false negatives (0%) indicates a strong bias in the model toward predicting the positive class.
- Low precision levels (0.31) indicate the model lacks reliability and should be optimized

Model Building II: Performance (Training Data)

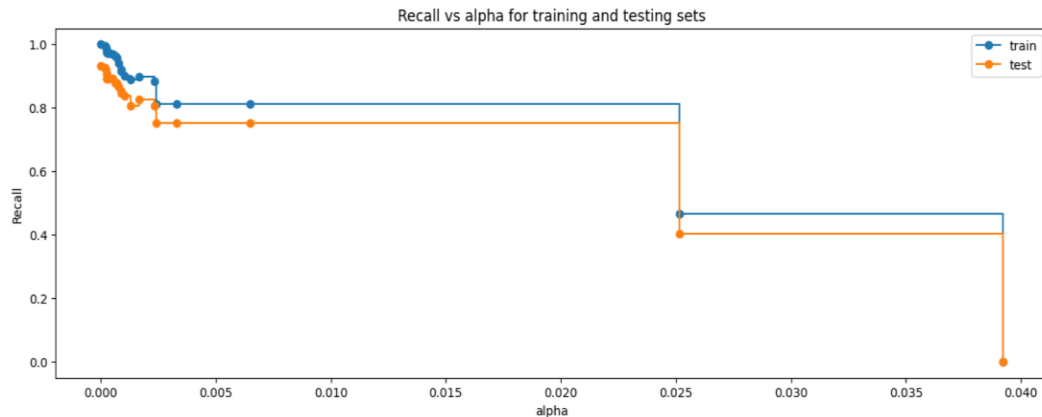
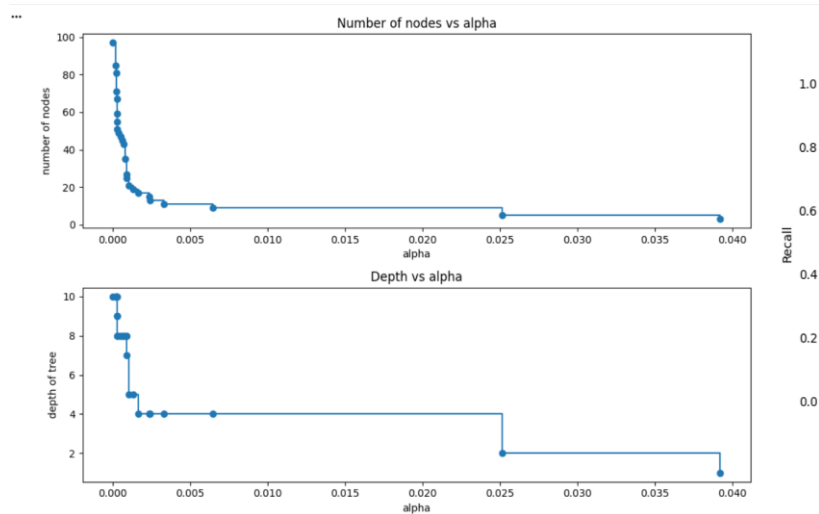
Decision Tree (Pre-Pruning)



- The decision tree is very shallow and strongly pruned
- Primary drivers include: (1) Income, (2) CCAvg, and (3) Family
- There is a high likelihood of underfitting
- Potential problems with shallow trees include:
 - (a) Low Training and Test Accuracy results
 - (b) Favor dominant patterns
 - (c) Highly sensitive to bias
 - (d) Fails to capture complex patterns
 - (e) Only models simple splits

Model Building III: Performance (Training Data)

Decision Tree (Post-Pruning)



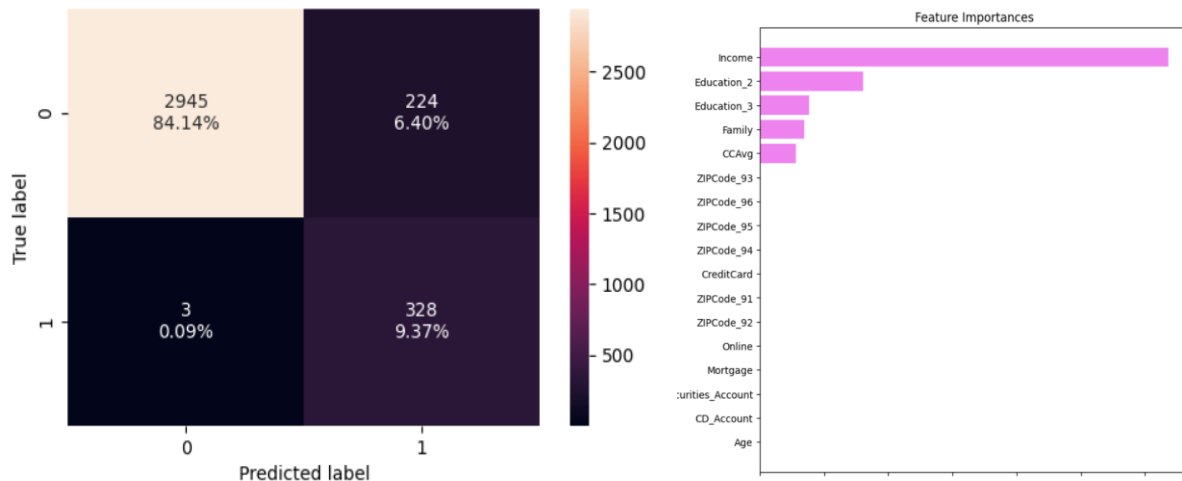
At low alpha values (0.000–0.002), the model remains highly complex with 40–100 nodes. Increasing alpha to 0.003–0.020 effectively prunes the tree to 10–15 nodes while preserving structure. At high alpha values (0.025–0.040), the tree becomes over-pruned, collapsing to just 1–5 nodes.

The alpha range of 0.003–0.020 delivers the best generalization, with recall consistently between 0.75 and 0.85. In contrast, alpha values from 0.025–0.040 result in a sharp decline in recall, indicating underfitting.

The most optimal `ccp_alpha` to optimize the model is ~0.010

Model Building III: Performance (Training Data)

Decision Tree (Post-Pruning)



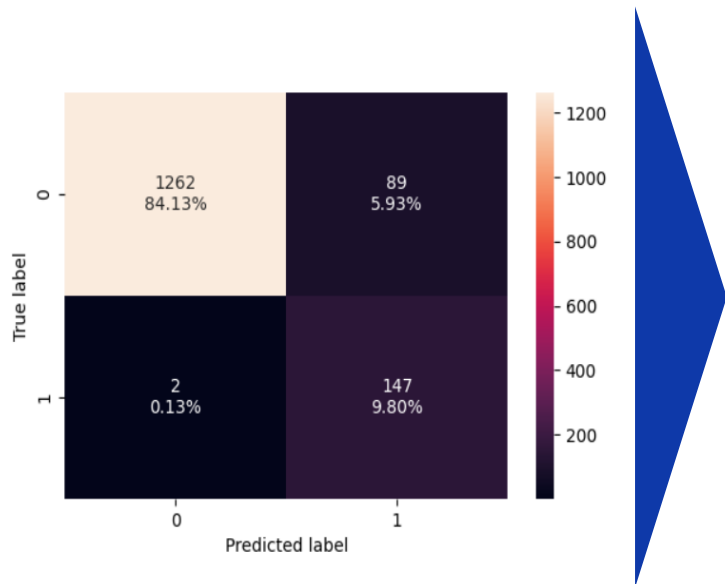
Performance Metrics

- **Accuracy: 0.93**
- **Recall: 0.99**
- **Precision: 0.59**
- **F1: 0.74**

- This is a high performing optimized model with very strong generalization with only 227 misclassifications out of 3500 observations.
- The model shows strong performance with **93% accuracy** and **99% recall**, effectively identifying positive cases. However, the **moderate precision (59%)** indicates false positives remain, and further tuning may improve the **F1-score (74%)** to better balance precision and recall.
- Key predictors of loan acceptance include: (1) Income, (2) Education Level (Graduate) and (2) Education Level (Advanced).

Model Building III: Performance (Test Data)

Decision Tree (Post-Pruning)



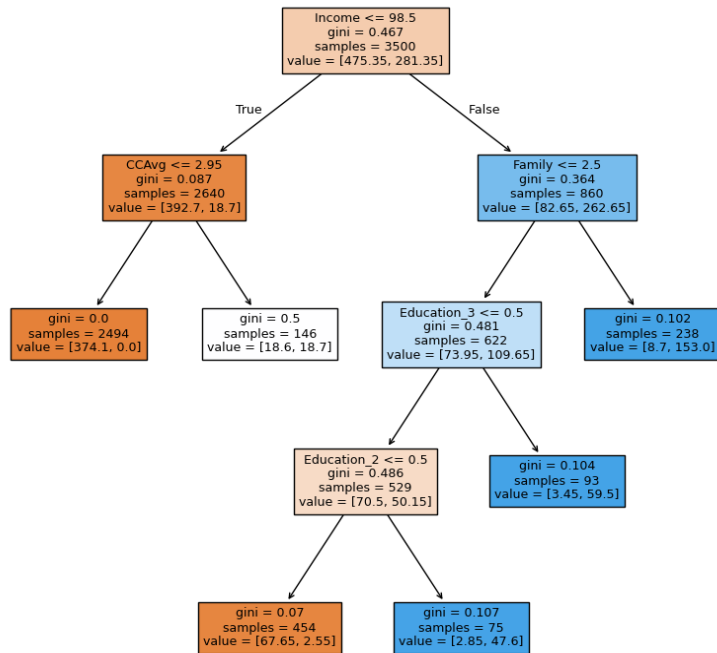
Performance Metrics

- **Accuracy: 0.93**
- **Recall: 0.98**
- **Precision: 0.62**
- **F1:0.76**

- This confusion matrix shows strong overall performance with only 95 misclassifications out of 1500 observations.
- The model shows strong performance with **93% accuracy** and **98% recall**, effectively identifying positive cases. However, the **moderate precision (62%)** indicates false positives remain while the **F1-score (76%)** indicates a solid balance between precision and recall.
- This represents a significant improvement in performance on unseen data compared to the training results, indicating better generalization.

Model Building III: Performance (Training Data)

Decision Tree (Post-Pruning)



- The decision tree is concise and interpretable
- Primary drivers include: (1) Income, (2) CCAvg, and (3) Family
- The tree is slightly heavier on the right side with Education Level (Advanced) and Education Level (Graduate).
- This tree is suitable for deployment if the focus is on recall and accuracy.
- Post deployment monitoring and retraining would be required to further optimize the model.

Model Performance Improvement

... Training performance comparison:

	Decision Tree (sklearn default)	Decision Tree (Pre-Pruning)	Best Performing Model Decision Tree (Post-Pruning)
Accuracy	1.0	0.790286	0.935143
Recall	1.0	1.000000	0.990937
Precision	1.0	0.310798	0.594203
F1	1.0	0.474212	0.742922

Test set performance comparison:

	Decision Tree (sklearn default)	Decision Tree (Pre-Pruning)	Decision Tree (Post-Pruning)
Accuracy	0.986000	0.779333	0.939333
Recall	0.932886	1.000000	0.986577
Precision	0.926667	0.310417	0.622881
F1	0.929766	0.473768	0.763636

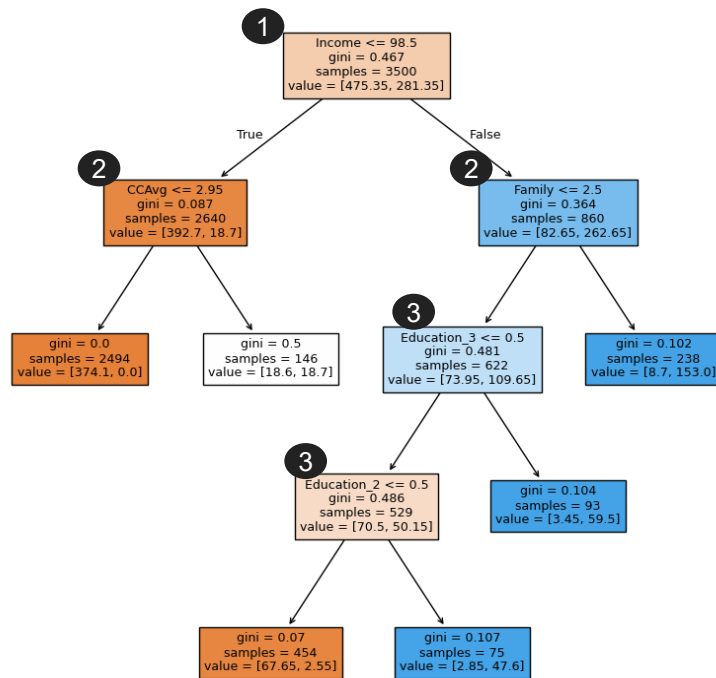
The **post-pruned decision tree** offers the best balance between predictive performance, generalization, and interpretability.

The **pre-pruned decision tree**, on the other hand, is overly restrictive. While it achieves perfect recall on both training and test sets, its extremely low precision results in weak accuracy and F1 scores.

The **default sklearn tree** achieves slightly higher test metrics, its perfect training performance indicates overfitting risk, making it less reliable for deployment.

Decision Rules and Feature Importance

Best Performing Model (Post Pruning)



Decision Rules

- 1 Income is the first split at ≤ 98.5 | gini: 0.467
 - Customers with higher income levels are more likely to accept a personal loan.
- 2 Credit Card Spending (CCAvg) ≤ 2.95 | gini: 0.087
 - Customers with higher CCAvg spending are more likely to accept a personal loan.
- 2 Family ≤ 2.5 | gini: 0.364
 - Large families show higher acceptance rates.
- 3 Education ≤ 0.5 | gini: 0.481
 - Customers with advanced degrees are more likely to accept a personal loan.

Feature Importance

- 1 Income: The most influential predictor, contributing the largest share of information gain
- 2 Credit Card Spending (CCAVG): The second most important feature, capturing customer spending behavior and credit usage patterns
- 2 Family Size: Adds incremental value by identifying customers with higher financial need
- 3 Education Level: A strong categorical driver, especially distinguishing advanced degrees from undergrad degree holders.



Happy Learning !

